SHERPA

Sports, Heritage and Excursion Recommendation and Planning Agent

An agentic, governed decision-support platform for sporting and cultural outings

Foundational Architecture Document

Enterprise Architecture (TOGAF 10) and Solution Architecture

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1. Executive Summary

SHERPA is a conversational, agentic decision-support platform that turns a natural-language description of a planned sporting or cultural outing into a complete, safety-aware activity dossier (the ficha técnica). The user describes the outing in their own words, or fills in a light template with a free-text field for additional characteristics, and the system determines what is needed, gathers reliable information from open public data, and produces a structured, reviewable document with recommendations.

The platform deliberately combines three complementary forms of intelligence. Generative AI is used for natural-language understanding and for drafting engaging, human-readable narrative. Classical (traditional) AI is used for the parts that must be deterministic, explainable and auditable, such as equipment recommendation, risk scoring and pattern detection over past outings. A governed multi-agent layer orchestrates the work, using Solace Agent Mesh as the event-driven backbone, so that each specialised agent (weather, geography and access, route, heritage, recommendation, dossier) can be observed, traced and governed. Databricks provides the data pipelines that ingest, clean and curate the open public data feeding the system.

A core design stance, carried over from earlier case studies in the wider portfolio, is that governance and human oversight are designed from day zero rather than added retroactively. Because the platform offers advice that can affect personal safety in outdoor environments, a Human-In-The-Loop (HITL) checkpoint is mandatory before the final dossier is issued. The reviewer can approve (generating a PDF), request modifications in free-form natural language, or take other actions described in this document. Every decision is recorded in an append-only, hash-chained audit trail.

The document also makes explicit a key constraint discovered during research: Wikiloc does not offer a public API. SHERPA therefore adopts a bring-your-own-GPX pattern complemented by open routing engines built on OpenStreetMap data, rather than relying on fragile and legally questionable scraping.

Two deployment tracks are described throughout. A Pilot or Demo track uses only free and open-source technologies, suitable for learning and for portfolio demonstration. A Production track provides four parallel cloud blueprints, on Amazon Web Services, Microsoft Azure, Google Cloud Platform and Alibaba Cloud, with the rationale for each choice. The architecture is intentionally vendor-neutral so that the same logical design can be realised on any of these platforms without rework.

This document covers the strategic and architectural foundations. It does not contain setup steps or code, which will be delivered separately as an Implementation Guide.

2. Case Study Definition

2.1 Problem statement and opportunity

Planning an outdoor or cultural outing is surprisingly fragmented. A person who wants to organise a Sunday hike, a club cycling ride, a padel match or a heritage walk must currently consult several disconnected sources: one or more weather services, a trail repository for a suitable route, scattered official notices to check whether a path or site is closed, and assorted blogs or encyclopaedias to learn about the place. They must then translate all of this into practical decisions about kit, timing, water, food and meeting points. The process is time-consuming, error-prone and, importantly, safety-relevant: people regularly set off under-equipped, in deteriorating weather, or onto trails that are closed or unsuitable for the group.

The opportunity is to consolidate trustworthy open data and proven recommendation logic into a single conversational assistant that produces a clear, reviewable dossier. The same engine serves casual individuals, club organisers with a duty of care towards members, and tourism operators who want to enrich and de-risk the visitor experience.

2.2 Vision

Vision: to be the trusted planning companion for sporting and cultural outings, transforming a few sentences of intent into a complete, safety-aware and beautifully informative activity dossier, with reliable open data, explainable recommendations and human oversight at its core.

2.3 Conversational flow and intake

SHERPA supports two intake modes that converge on the same internal representation. The first is a fully conversational mode in which the user describes the outing in natural language and the assistant elicits anything missing through dialogue. The second is a structured template that captures the core fields quickly, complemented by an Other characteristics free-text field for nuances such as group composition, fitness level, presence of children or dogs, dietary needs, or a preference for shade. In both modes a Generative AI layer interprets the input and a validation step confirms completeness before processing begins.

2.3.1 Required inputs

Input	Description and derived data	Validation and notes
Activity type	Cycling (road, gravel, mountain bike), hiking and trekking, trail running, tennis, padel, climbing, kayaking and paddle, cultural outing (museum, monument, festival, urban walk), and others	Drives the recommendation logic and the equipment checklist; constrained to a managed taxonomy
Departure origin	The home address from which the group sets off; geocoded to coordinates	Personal data; minimised, encrypted and used only to compute travel to the start
Activity start point	The start of the track (typically the first point of a Wikiloc or equivalent GPX file)	Coordinates; used to fetch local weather, access and heritage

Input	Description and derived data	Validation and notes
		context
Activity end point	The destination point of the same track	Used to detect loop versus point-to-point; informs logistics and return
Date	The date of the activity	Determines seasonality and the forecast horizon (see assumptions)
Start time	Planned start time	Drives hourly weather, daylight and timing of the schedule
End time	Planned end time	With start time yields duration, a key driver of kit, water and food recommendations

From these inputs the system derives further attributes used throughout: distance and cumulative elevation gain (from the GPX track), estimated duration where not supplied, region and municipality (for the relevant weather and authority data), and season. The departure origin combined with the start point yields the approach logistics and a sensible meeting point.

2.4 Processing Capabilities

2.4.1 Meteorology

The system retrieves weather for the day and location of the activity from open meteorological services rather than by scraping. For Spain, AEMET OpenData provides a free REST API (registration required for an API key) with municipal daily and hourly forecasts and official warnings (avisos). For global coverage and for historical baselines, Open-Meteo provides a free JSON API that requires no key for non-commercial use, including forecast and ERA5 reanalysis data, an elevation service and air-quality data. From the raw variables the system derives comfort and risk indicators: temperature with wind-chill and heat-index adjustments, wind speed and gusts, probability and intensity of rain, snow, ultraviolet exposure and thunderstorm risk.

Important limitation: numerical forecasts are reliable only for roughly the next seven to ten days. For activities planned further ahead, the system presents a climatological expectation derived from ERA5 reanalysis, clearly labelled as a seasonal baseline rather than a forecast, and schedules a re-check closer to the date.

2.4.2 Place intelligence and access

The system investigates whether part or all of the route may be closed or restricted on the chosen day, and surfaces other factors that may affect the outing positively or negatively. Authoritative closure data is fragmented in Spain and elsewhere, so the design combines several sources and is explicit about uncertainty. OpenStreetMap, queried through the Overpass API, supplies access tags, barriers, surface, seasonal restrictions and points of interest. Official protected-area sources (the national parks network and the regional administrations) are consulted where structured data exists. Local event calendars help detect days when a site is congested or closed for an event. When the evidence is incomplete, the agent flags the uncertainty and recommends confirming with the official source, rather than asserting a closure it cannot verify.

2.4.3 Route and track sourcing

Research finding that shapes the design: Wikiloc does not provide a public API, by its own statement, for technical and legal or privacy reasons. SHERPA therefore does not attempt to scrape Wikiloc, which would be fragile and contrary to its terms.

Three robust patterns are adopted instead. The primary pattern is bring-your-own-GPX: the user supplies the GPX file they have already downloaded from their own Wikiloc account or recorded on their device, and the system parses and enriches it. The second pattern uses open routing engines built on OpenStreetMap data to generate or complete a route between the supplied start and end points, with profiles for hiking, cycling and other modes, and with elevation. Suitable free engines include OpenRouteService (free API key), GraphHopper, OSRM, BRouter and Valhalla. The third, optional pattern integrates third-party platforms such as Strava or Komoot through their official APIs where licensing and OAuth permit, for users who prefer to connect those accounts.

2.4.4 Cultural and natural enrichment

To add the beautiful details that make an outing memorable, the system gathers context on the history of the place, its landscapes, and its likely flora and fauna. Wikipedia (via its REST API) and Wikidata (via SPARQL) provide historical and points-of-interest context with provenance. Wikimedia Commons supplies imagery with licence awareness. The Global Biodiversity Information Facility (GBIF) and iNaturalist provide records of species observed in the area and season. Regional heritage open-data portals add monuments and cultural assets. A Generative AI step synthesises this material into engaging narrative, grounded in the retrieved sources and citing them to reduce the risk of fabrication.

2.4.5 Recommendations through traditional AI

Recommendations are produced primarily by classical AI rather than by the language model. A combination of rules and trained models maps the activity type, distance, elevation, duration, weather and season to an equipment checklist and a set of risk flags. Similarity and clustering techniques compare the planned outing with historical ones to surface useful patterns, for example that comparable rides in similar heat benefited from an extra water bottle. Anomaly detection raises warnings when a combination is risky, such as a long exposed route in high heat with insufficient planned water.

Why classical AI for this layer: safety-relevant recommendations must be deterministic, explainable, auditable, inexpensive to run and straightforward to validate against regulation. The language model is therefore confined to understanding input and to writing prose, while the scoring that influences safety advice is handled by transparent models whose behaviour can be tested and reproduced.

2.4.6 Dossier (ficha técnica) composition

The dossier brings everything together into a single structured document. It contains the supplied data, the derived route information (track, distance, elevation profile, loop or point-to-point, and suggested meeting points), the schedule, a weather panel with risk indicators, an equipment checklist tailored to the activity, the place-intelligence and access notes, the cultural and natural highlights, and a clearly worded safety section with disclaimers.

Activity	Representative equipment logic (illustrative)
Cycling	Personal kit (helmet, layers for the forecast, sun protection, hydration and nutrition for the duration) and a bicycle pre-ride check (tyres and pressures, brakes, drivetrain, spares, tools, lights)
Hiking and trekking	Backpack sizing, water and food for duration and heat, navigation, layers and waterproofs, footwear, first aid, headtorch for late finishes
Tennis and padel	Rackets, suitable balls, court surface and footwear, sun protection, hydration; court booking or availability where relevant
Cultural outing	Tickets or booking, opening hours, comfortable footwear, weather-appropriate clothing, and timing around closures or events

2.5 Human-In-The-Loop review and decisions

Before any dossier is issued, the draft is presented to a human reviewer at a mandatory checkpoint. This is both a quality gate and a safety control. The reviewer can take the following actions, each of which is recorded in the audit trail:

1. Approve: generate the final PDF dossier, with optional export of the GPX track and an optional calendar invitation.
2. Modify: provide a free-text instruction in natural language describing what to change; the system re-runs only the affected parts and presents a new draft for re-review.
3. Regenerate a section: refresh a single element, such as the weather panel or the cultural highlights, without rebuilding the whole dossier.
4. Save as draft: persist the work and resume later.
5. Reject or discard: abandon the draft, with a recorded reason.
6. Acknowledge and escalate: when a severe weather warning or an access closure is detected, require an explicit acknowledgement before approval is allowed.
7. Share: distribute the dossier to the group or generate a shareable link.
8. Schedule a reminder: automatically re-check the weather, for example twenty-four hours before the activity, and notify the organiser if conditions change materially.

2.6 Personas

The platform is designed around several recurring users: the weekend hiker planning a family walk; the cycling club organiser arranging a group ride with a duty of care; the padel or tennis group coordinator; the cultural-tourism guide preparing an enriched visit; the club or federation secretary who must document outings; and the family planner balancing children, dogs and varied fitness levels.

2.7 Assumptions and Constraints

- Coverage: meteorological detail is richest for Spain through AEMET; global coverage is provided through Open-Meteo, with the understanding that some national sources are more authoritative locally.
- Forecast horizon: numerical forecasts are trustworthy for about a week; beyond that the system uses clearly labelled climatological baselines.
- Wikiloc has no public API; the design relies on user-supplied GPX and open routing engines rather than scraping.
- Advisory nature: SHERPA provides advice, not authoritative safety guarantees; the user remains responsible for their decisions, and this is stated in the dossier.
- Licensing: open-data licences are respected throughout, including AEMET reuse under Spanish Law 18/2015, OpenStreetMap attribution under the ODbL, Wikimedia content licences, and GBIF citation requirements.
- Privacy: the home address is personal data and is subject to data minimisation, encryption and retention limits under the GDPR.

3. Commercial Approach

This section frames why an organisation would want to implement SHERPA. It is written to be persuasive to prospective adopters while remaining honest about scope. It is not financial advice, and the figures discussed are qualitative illustrations of value rather than guarantees.

3.1 Value Proposition

SHERPA converts fragmented, manual outing planning into a single conversational step that yields a trustworthy, reviewable dossier. It reduces planning time, improves safety through explainable risk-aware recommendations, and enriches the experience with curated heritage and natural context. Because it is built on open public data and a vendor-neutral architecture, the marginal cost of each plan is low and the platform avoids lock-in.

3.2 Target Segments and the Benefit to each

Segment	Why they adopt	How they benefit
Sports clubs and federations	Duty of care and member engagement	Documented, safety-aware ride and walk plans; fewer incidents; stronger community
Tourism boards and destination management	Enrich and distribute visitors	Heritage-rich itineraries that spread footfall and showcase lesser-known sites
Outdoor retailers	Contextual, trusted recommendations	Relevant kit suggestions at the moment of planning, a natural and non-intrusive conversion point
Insurers and mutual societies	Loss prevention	Risk-aware planning lowers the frequency of avoidable incidents among members
Guides and micro-enterprises	Productivity	Automatic generation of professional dossiers for clients in minutes
Schools, scouts and youth groups	Safe educational outings	Age-appropriate, well-prepared excursions with clear checklists

3.3 Monetisation Options

Several models can be combined. A freemium consumer tier introduces individuals to the service. A business-to-business subscription serves clubs, guides and tourism operators on a per-seat or per-organisation basis. A white-label offering lets destination management organisations brand the experience. An affiliate model on equipment recommendations creates a value-aligned commerce stream. Aggregated, privacy-preserving insight products can serve planners and authorities. Finally, an API offering exposes the planning capability to partner applications.

3.4 Differentiators

- Governed agentic architecture: every recommendation is traceable and auditable, which matters for clubs and insurers.

- Safety-first responsible AI: classical models drive safety scoring, with mandatory human review before issuance.
- Open-data backbone: low marginal cost and transparent provenance.
- Vendor-neutral and portable: the same design runs on any major cloud, protecting the adopter from lock-in.
- Modular verticals: an adopter can start with a single activity, such as hiking, and expand.

3.5 Indicative Business Case

The qualitative return rests on three levers. First, time saved: a plan that previously took thirty to sixty minutes across several tools is produced in minutes. Second, risk reduction: better-prepared groups experience fewer weather and equipment related incidents, which is directly valuable to clubs and insurers. Third, engagement and reach: enriched, shareable dossiers increase member satisfaction and, for tourism bodies, help distribute visitors across a wider set of sites. Adopters can pilot the platform on free, open-source technology before committing to a production cloud, which keeps the initial investment low and the learning fast.

4. Enterprise Architecture (TOGAF 10)

This section presents the enterprise architecture using the TOGAF 10 Architecture Development Method (ADM) as its frame, covering the Architecture Vision and the Business, Data, Application and Technology domains, together with the cross-cutting concerns of Requirements Management and Architecture Governance. ArchiMate 3.2 is the modelling language of reference and DAMA DMBOK2 informs the data governance perspective. The intent is a professional architecture grounded in real experience of both enterprise systems and the organisation of sporting and cultural activities, not merely a list of components.

4.1 Architecture Vision (ADM Phase A)

The vision establishes scope, stakeholders and the principles that constrain every later decision. SHERPA exists to turn intent into a safe, enriched and reviewable plan. Its success is measured by the quality and safety of the dossiers it produces, the trust of organisers and members, and the ease with which the same design moves from a free pilot to any production cloud.

4.1.1 Stakeholders and concerns

Stakeholder	Primary concerns
Activity organisers and members	Safety, clarity, time saved, an enjoyable and well-prepared outing
Clubs, federations and guides	Duty of care, documentation, repeatability, reputation
Tourism authorities	Visitor experience, heritage promotion, sustainable distribution of footfall
Data protection and compliance officers	Lawful handling of personal data, transparency, the EU AI Act posture
Platform operators	Reliability, observability, cost control, portability across clouds
Architects and engineers	Modularity, reuse, testability, clean governance and clear standards

4.1.2 Architecture Principles

Principle	Statement and rationale	Implication
Governance by design	Governance and audit are built in from day zero because retroactive governance is architecturally impossible	Hash-chained audit trail; policy enforced at the orchestration layer
Human accountability for safety	A human approves any dossier that carries safety implications	Mandatory HITL checkpoint before issuance; explicit acknowledgement of warnings
Explainable and traditional AI for decisions	Safety-relevant scoring uses transparent models, not opaque generation	Classical AI for recommendations; the language model only understands and drafts

Principle	Statement and rationale	Implication
Open data first	Prefer authoritative open data over scraping	AEMET, Open-Meteo, OpenStreetMap, Wikidata, GBIF; no Wikiloc scraping
Cloud portability and no lock-in	The logical design is vendor-neutral	Event-driven mesh and open formats; four production blueprints
Privacy by design and default	Personal data is minimised and protected	Encrypt and minimise the home address; clear retention limits
Reuse over rebuild	Leverage lessons and assets from earlier case studies	Reuse patterns, API keys via secrets manager, and proven components
Cost-conscious pilot	Demonstrate value on free and open-source technology first	A complete pilot at zero licensing cost before any cloud commitment

4.2 Business Architecture (ADM Phase B)

The business architecture describes what the platform does in capability terms, independent of technology. The principal capabilities are: Activity Intake and Understanding; Data Acquisition and Curation; Risk Assessment and Recommendation; Dossier Generation; Human Review and Governance; Distribution and Notification; and Identity and Account Management.

4.2.1 Primary value stream: plan an outing

The end-to-end value stream proceeds from intent to issued dossier: capture intent in natural language, complete and validate the request, acquire weather, access, route and heritage data, assess risk and generate recommendations through classical AI, compose the draft dossier, submit it for human review, and, on approval, issue the PDF and optional artefacts and schedule any reminders. Each stage maps to one or more business capabilities and, later, to application services and agents.

4.2.2 Business services and roles

Business service	Description
Outing planning	Accepts intent and returns a complete, reviewable dossier
Safety advisory	Produces explainable risk indicators and an acknowledgement workflow
Heritage enrichment	Adds curated historical, landscape and biodiversity context
Plan governance	Records decisions, enforces review, and maintains the audit trail
Distribution	Issues the PDF, shares plans and schedules reminders

Key roles in the operating model include the Organiser (the human reviewer at the HITL gate), the Platform Operator (reliability and cost), the Data Steward (quality, provenance and licensing) and the Architecture Governance owner (principles, standards and the change process).

4.3 Data Architecture (ADM Phase C, Data)

The data architecture defines the principal information entities, how data flows and is curated, and how it is governed. It aligns with DAMA DMBOK2 and uses a Medallion approach on Databricks for ingested open data.

4.3.1 Conceptual entities

Entity	Description
User and Household	The organiser and the protected home address used for logistics
Outing (Plan)	The planned activity instance with its inputs, status and decisions
Activity Type	Reference taxonomy that drives recommendations and checklists
Location and Track	Start, end and the GPX geometry, with derived distance and elevation
Weather Forecast and Observation	Forecast and historical climatology for the location and date
Place Fact	History, points of interest, access and closure information
Biodiversity Record	Likely flora and fauna for the area and season
Equipment Item and Checklist	Reference catalogue and the generated, tailored checklist
Recommendation and Risk Flag	The explainable outputs of the classical models
Dossier	The composed ficha técnica and its issued PDF
Audit Event	An append-only, hash-chained record of every decision

4.3.2 Curation, Governance and Provenance

Ingested open data flows through a Medallion pipeline: a bronze layer captures raw responses with their source and timestamp; a silver layer standardises units, geographies and schemas; and a gold layer exposes curated, query-ready datasets to the agents and models. Every record carries provenance and licence metadata so that attribution and reuse obligations are honoured. Data quality is monitored, lineage is tracked, and personal data is segregated, encrypted and subject to explicit retention rules. A lesson from earlier work is applied here: schema discipline, with explicit column lists on writes and merges, prevents schema contamination across runs.

4.4 Application Architecture (ADM Phase C, Application)

The application architecture describes the logical components and how they communicate. Components are loosely coupled and communicate through events on the Solace mesh, with agent-to-agent (A2A) messaging for delegation and Model Context Protocol (MCP) tools wrapping the external data services. A conversational interface and REST gateways provide the human-facing and integration surfaces.

Application component	Responsibility
Conversational interface	Captures intent, runs the dialogue, and presents the HITL review console
Intake and validation service	Normalises input, fills gaps, and produces the internal request representation
Orchestration layer	Coordinates the specialised agents and enforces governance policy
Specialised agents	Weather, geography and access, route, heritage, recommendation, and dossier agents
Recommendation engine	Hosts the classical models and rules for equipment and risk
Dossier and PDF service	Composes the ficha técnica and renders the final document
Notification service	Sends shares, reminders and acknowledgement requests
Identity and audit services	Authenticate users and maintain the hash-chained audit trail

4.5 Technology Architecture (ADM Phase D)

The technology architecture provides the platform building blocks that realise the application and data designs: compute and container runtimes, the event broker, the data platform, the vector and graph stores, object storage, the language-model serving layer, observability, continuous integration and delivery, and secrets management. The detailed technology selections, for both the pilot and the four production clouds, are given in Section 5. The guiding standard is that every building block has an open-source realisation for the pilot and a managed equivalent on each production cloud, preserving portability.

4.6 Architecture Governance and Requirements Management

Requirements Management sits at the centre of the ADM and keeps the architecture aligned with stakeholder needs as they evolve. Architecture Governance ensures that the principles are honoured. Decisions are captured as Architecture Decision Records (ADRs) in the repository, a lightweight governance review approves significant changes, and compliance is enforced through the HITL gate and the audit trail.

4.6.1 Regulatory Landscape

Area	Posture
GDPR	The home address is personal data; minimise, encrypt and retain only as needed; provide transparency and rights handling
EU AI Act	Most functions are limited-risk, requiring transparency; because the platform offers safety-relevant advice, human oversight and clear disclaimers are applied as proportionate mitigations
Digital Services Act	Content enrichment respects provenance and licensing; shared content carries attribution
Open-data licences	AEMET reuse under Law 18/2015; OpenStreetMap under the ODbL; Wikimedia content licences; GBIF citation
Accessibility	User interfaces target WCAG and EN 301 549 conformance

4.6.2 Principal Risks

- Forecast uncertainty leading to unsafe plans; mitigated by labelled baselines, re-check reminders and human acknowledgement of warnings.
- Incomplete or stale closure data; mitigated by multi-source consultation and explicit uncertainty flags.
- Over-reliance on generated narrative; mitigated by grounding in cited sources and by confining safety scoring to classical models.
- Personal-data exposure; mitigated by minimisation, encryption and strict retention.
- Dependency and supply-chain fragility; mitigated by version pinning and vendor-neutral design.

5. Solution Architecture

This section specifies the solution across six layers: Application, Data, AI, Agentic AI, Technology (IT) and Security. For each layer it states what is used, how and why. Two technology stacks are given throughout. The Pilot or Demo stack uses only free and open-source technologies and is the basis for learning and portfolio demonstration. The Production stack provides four parallel blueprints, on AWS, Azure, GCP and Alibaba Cloud. This section deliberately contains no setup steps or code, which will follow in a separate Implementation Guide.

5.1 Application Architecture

What: a conversational front end, a set of back-end services, and a HITL review console. How: the front end captures intent and renders the review console; back-end services are event-driven and stateless where possible, communicating through the Solace mesh; the dossier service renders a PDF. Why: a conversational surface lowers the barrier to planning, while event-driven decoupling gives resilience, observability and independent scaling.

Layer Concern	Pilot (open source, free)	AWS	Azure	GCP	Alibaba Cloud
Conversational and review UI	Streamlit	Amplify or ECS Fargate	App Service or Static Web Apps	Cloud Run or Firebase Hosting	Serverless App Engine with OSS UI
API gateway	FastAPI	API Gateway	API Management	API Gateway or Apigee	API Gateway
PDF rendering	WeasyPrint or ReportLab	Container on ECS	Container on Container Apps	Container on Cloud Run	Container on SAE

5.2 Data Architecture

What: pipelines that ingest, curate and serve open data, plus stores for vectors and relationships. How: Databricks with Delta Lake runs the Medallion pipelines (bronze, silver, gold); a vector store powers retrieval over place facts; a graph store captures relationships among places, tracks, points of interest and species. Why: the Medallion pattern gives clean, governed, query-ready data with full lineage; vector retrieval grounds the narrative; the graph reveals connections that enrich recommendations.

Layer Concern	Pilot (open source, free)	AWS	Azure	GCP	Alibaba Cloud
Data platform and pipelines	Databricks Community or local Spark with Delta	Databricks on AWS	Azure Databricks	Databricks on GCP	E-MapReduce or MaxCompute with Delta
Object storage	MinIO or local volume	Amazon S3	ADLS Gen2 or Blob	Cloud Storage	Object Storage Service
Vector store	Qdrant (Docker)	Qdrant on EKS or OpenSearch	Azure AI Search or pgvector	Vertex Vector Search or AlloyDB pgvector	AnalyticDB for PostgreSQL or Hologres
Graph store	Neo4j Community with APOC	Neo4j Aura or Neptune	Neo4j Aura or Cosmos DB Gremlin	Neo4j Aura on GCP	Neo4j self-managed or GraphDB
Relational and metadata	PostgreSQL	Amazon RDS or Aurora	Azure Database for PostgreSQL	Cloud SQL	ApsaraDB RDS

Lessons reused here include pinning the Qdrant client to the server minor version and using the current query interface, setting the Neo4j enhanced schema flag off and enabling APOC, embedding text with the nomic-embed-text model at 768 dimensions in-process, and enforcing explicit column lists on Delta writes and merges to avoid schema contamination.

5.3 AI Architecture (Classical and Traditional AI)

What: the deterministic intelligence that drives safety-relevant recommendations. How: a feature pipeline derives attributes from the inputs and weather; rules and trained models (for example gradient-boosted trees) produce the equipment checklist and risk flags; experiment tracking and a model registry manage versions; inference runs in batch for planning and online for interactive use. Why: this layer must be explainable, testable, reproducible and inexpensive, which classical models provide far better than generation.

Layer Concern	Pilot (open source, free)	AWS	Azure	GCP	Alibaba Cloud
Modelling library	scikit-learn and XGBoost	Same in SageMaker	Same in Azure ML	Same in Vertex AI	Same in PAI
Experiment tracking and registry	MLflow	SageMaker with MLflow	Azure ML registry	Vertex AI Model Registry	PAI model registry
Feature management	Feature tables in Delta	SageMaker Feature Store	Azure ML feature store	Vertex AI Feature Store	PAI feature store
Serving	FastAPI microservice	SageMaker endpoints	Azure ML endpoints	Vertex AI endpoints	PAI-EAS

5.4 Agentic AI Architecture

What: the governed, event-driven multi-agent layer that orchestrates the work. How: Solace Agent Mesh provides the backbone, where specialised agents communicate over the Solace event broker using the agent-to-agent (A2A) protocol, an orchestrator decomposes the request and delegates, and MCP tools wrap the external data services. The language models perform understanding and drafting; the recommendation agent calls the classical models rather than reasoning about safety itself. Within an agent that requires a stateful pause, LangGraph provides the HITL pattern. Why: Solace Agent Mesh is open source, vendor-neutral and production-grade, gives complete observability because all communication flows through the broker, and scales cleanly from pilot to production, which directly serves the governance and portability principles.

5.4.1 Agent Catalogue

Agent	Responsibility
Orchestrator	Interprets the request, plans the work and delegates to specialists
Intake and understanding	Natural-language understanding, gap-filling and validation
Weather	Retrieves forecasts and baselines and derives comfort and risk indicators
Geography and access	Resolves locations and investigates closures and access restrictions
Route	Parses supplied GPX or generates a route through open routing engines
Heritage and nature	Gathers and grounds historical, landscape and biodiversity context
Recommendation	Calls the classical models to produce the checklist and risk flags
Dossier	Composes the ficha técnica and prepares it for review and rendering
Governance and HITL	Enforces the review gate and writes the audit trail

Layer Concern	Pilot (open source, free)	AWS	Azure	GCP	Alibaba Cloud
Agent framework and mesh	Solace Agent Mesh Community with PubSub+ (Docker)	Solace on AWS via Solace Cloud	Solace on Azure via Solace Cloud	Solace on GCP via Solace Cloud	Solace self-managed on Alibaba
In-agent stateful HITL	LangGraph with a MemorySaver checkpointer	Same, containerised	Same, containerised	Same, containerised	Same, containerised
Crew-style sub-tasks	CrewAI where helpful	Same	Same	Same	Same
Language models	Groq API (llama-3.3-70b primary, 8b fallback); Ollama qwen2.5:3b local	Amazon Bedrock	Azure OpenAI or AI Foundry	Vertex AI (Gemini)	Model Studio (Qwen) or PAI-EAS

Lessons reused here include using the interrupt-before pattern with a module-level checkpointer singleton to enable workflow resumption after review, importing the crew framework in the main thread before spawning workers, preferring a local model that supports function calling, and mitigating free-tier token limits through heuristic pre-filtering, serialised workers and pacing.

5.5 Technology (IT) Architecture

What: the runtime, networking, observability and delivery foundations. How: containerised services with an orchestrator, infrastructure as code, centralised observability and a disciplined delivery pipeline. Why: consistency between pilot and production, reproducibility, and operational visibility.

Layer Concern	Pilot (open source, free)	AWS	Azure	GCP	Alibaba Cloud
Runtime and Containers	Docker with uv-managed Python	ECS Fargate or EKS	Container Apps or AKS	Cloud Run or GKE	SAE or ACK
Infrastructure as Code	Terraform	Terraform or CDK	Terraform or Bicep	Terraform	Terraform or ROS
Observability	LangSmith with Prometheus, Grafana and OpenTelemetry	CloudWatch and X-Ray	Azure Monitor and App Insights	Cloud Monitoring and Trace	CloudMonitor and ARMS
CI and Code Quality	GitHub Actions with ruff	CodePipeline	Azure DevOps or GitHub	Cloud Build	Yunxiao or Flow

5.6 Security Architecture

What: identity, secrets, encryption, data protection and AI-specific safeguards. How: federated identity with OpenID Connect; all secrets in a managed vault, never hardcoded, reusing existing project keys through the secrets manager; encryption in transit and at rest; least-privilege access; guardrails on agent tool use to limit prompt-injection impact; content provenance to support trust; pinned dependencies to reduce supply-chain risk; and the hash-chained audit trail as tamper-evidence. Why: the platform handles personal data and offers safety-relevant advice, so security and integrity are first-order concerns.

Layer Concern	Pilot (open source, free)	AWS	Azure	GCP	Alibaba Cloud
Identity	Keycloak	Cognito	Microsoft Entra ID	Identity Platform	IDaaS and RAM
Secrets management	Local vault or dotenv-vault	Secrets Manager	Key Vault	Secret Manager	KMS and Secrets Manager
Encryption and keys	OpenSSL and disk encryption	KMS	Key Vault keys	Cloud KMS	KMS
Edge and protection	Reverse proxy with rate limiting	WAF and CloudFront	Front Door and WAF	Cloud Armor	WAF and CDN

5.6.1 Threat Model Summary

A STRIDE-style review highlights the protections that matter most here: spoofing is countered by federated identity; tampering by the hash-chained audit trail and signed artefacts; repudiation by comprehensive, attributable logging; information disclosure by minimisation and encryption of the home address and other personal data; denial of service by rate limiting and autoscaling; and elevation of privilege by least-privilege roles and scoped agent tools. Prompt injection is treated as a first-class risk, mitigated by constraining what tools each agent may call and by validating tool outputs before they influence safety advice.

6. Next Steps

The immediate path forward, following the strict approval-before-proceeding methodology used across the portfolio, is as follows.

1. Review and approve this foundational document, or request modifications, at the HITL checkpoint.
2. Confirm the project name SHERPA, or propose an alternative.
3. Create the GitHub repository with the GitHub CLI, public, with a Python gitignore and the Apache 2.0 licence.
4. Select the first activity vertical for the minimum viable product; hiking is recommended because it has the richest open data and the clearest safety value.
5. Confirm reuse of existing API keys (Groq, AEMET, OpenRouteService, Neo4j Aura) through the secrets manager, never hardcoded.
6. Commission the next deliverable, the Implementation Guide for the pilot, which will contain the setup and code.
7. Agree a day-by-day build plan in the style of the bootcamp, with deploy, test, report, fix and redeploy iterations.

6.1 Suggested Repository Creation Command

```
gh repo create mgalanme/sherpa --public --description "SHERPA: Sports, Heritage and Excursion Recommendation and Planning Agent" --gitignore Python --license apache-2.0
```

A first-pass repository structure could separate documentation, the pilot application, the agents, the data pipelines, the classical models and the infrastructure, mirroring the six solution layers in this document, so that the codebase reads as a direct realisation of the architecture.

Appendix A. Open Data Sources Catalogue

Source	Data provided	Access and licence
AEMET OpenData	Spain weather: municipal daily and hourly forecasts, observations, official warnings	Free REST API with a registration key; reuse under Spanish Law 18/2015
Open-Meteo	Global forecast, ERA5 historical reanalysis, elevation, air quality	Free JSON API, no key for non-commercial use
OpenStreetMap (Overpass)	Access tags, barriers, surfaces, points of interest, seasonal restrictions	Open data under the ODbL; attribution required
OpenRouteService, GraphHopper, OSRM, BRouter, Valhalla	Routing and elevation for hiking, cycling and other profiles	Free tiers or self-hosted; built on OpenStreetMap
Wikipedia REST and Wikidata SPARQL	History, points of interest and structured facts	Content under Creative Commons; attribution per licence
Wikimedia Commons	Imagery for the dossier	Per-file Creative Commons or public-domain licences
GBIF and iNaturalist	Biodiversity records for likely flora and fauna	Open with citation requirements

Note on Wikiloc: Wikiloc has stated that it does not offer a public API for technical and legal or privacy reasons. SHERPA therefore relies on user-supplied GPX files and on open routing engines, and does not scrape Wikiloc.

Appendix B. Glossary

Term	Meaning
Ficha técnica	The structured activity dossier produced by the platform
HITL	Human-In-The-Loop, the mandatory human review and decision checkpoint
A2A	Agent-to-Agent protocol used for delegation between agents
MCP	Model Context Protocol, used to expose external tools to agents
Medallion	The bronze, silver and gold layering of curated data
GPX	GPS Exchange Format, the open XML format for tracks and waypoints
ADM	The TOGAF Architecture Development Method
ADR	Architecture Decision Record

End of document. *This is a draft for review. On approval, the next deliverable will be the pilot Implementation Guide containing setup and code.*